

Theory Reference

Theoretical Foundations of Squeezed Light and Optical Parametric Amplification

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This document develops the theoretical foundations underlying squeezed light generation in an optical parametric amplifier. Starting from the quantized electromagnetic field, we build up the formalism of coherent and squeezed states, derive the $\chi^{(2)}$ nonlinear interactions governing photon pair creation, establish the phase-matching conditions required for efficient parametric interaction, and treat the corrections arising from focused Gaussian beams via Boyd–Kleinman theory. The optical parametric amplifier is then developed as a resonantly enhanced implementation of the single-mode squeezing operator, with its algebraic structure rooted in the group $SU(1,1)$, culminating in the Collett–Gardiner input–output result for the squeezed quadrature variance. The treatment is self-contained and intended as a reference for the companion design document.

1 Quantizing the Electromagnetic Field

TO understand squeezed light we must first understand the mathematical language of light itself. To do this we quantize the electromagnetic field. Consider a single radiation mode confined to a one-dimensional cavity of length L .

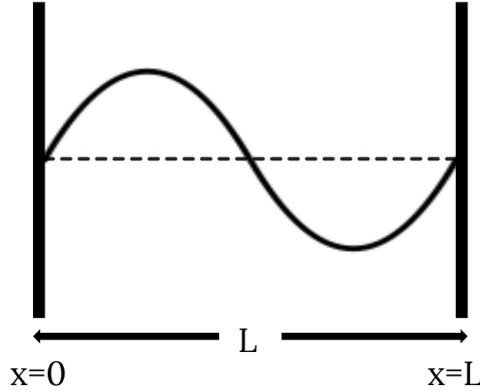


Figure 1: A single $n = 1$ standing-wave mode confined between 1D cavity walls at $x = 0$ and $x = L$. The field vanishes at both boundaries.

The field vanishes at the boundaries $x = 0$ and $x = L$ and is therefore a standing wave. Maxwell’s equations in the absence of sources are

$$\begin{aligned} \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}, \\ \nabla \cdot \mathbf{E} &= 0, & \nabla \cdot \mathbf{B} &= 0. \end{aligned} \tag{1}$$

The energy stored in the cavity can be written as

$$H = \frac{1}{2} \int dV \left[\varepsilon_0 E^2(\mathbf{r}, t) + \frac{1}{\mu_0} B^2(\mathbf{r}, t) \right] = \frac{1}{2} (p^2 + \omega^2 q^2), \quad (2)$$

which is precisely a harmonic oscillator with canonical variables q (related to E) and $p \propto \dot{q}$ (related to B).

Quantization proceeds by promoting q and p to operators satisfying $[\hat{q}, \hat{p}] = i\hbar$, and introducing the photon annihilation and creation operators

$$\hat{a} = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} + i\hat{p}), \quad \hat{a}^\dagger = \frac{1}{\sqrt{2\hbar\omega}}(\omega\hat{q} - i\hat{p}), \quad (3)$$

satisfying $[\hat{a}, \hat{a}^\dagger] = 1$. The Hamiltonian becomes the familiar $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$. The number (Fock) states $|n\rangle$ are eigenstates of $\hat{n} = \hat{a}^\dagger\hat{a}$, generated by repeated application of $\hat{a}^\dagger$ on the vacuum:

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle. \quad (4)$$

Because $\hat{n}$ does not commute with the field operator, a state of well-defined photon number has a completely uncertain field: $\langle n|\hat{E}|n\rangle = 0$ while $\langle n|\hat{E}^2|n\rangle \neq 0$, reflecting irreducible quantum fluctuations.

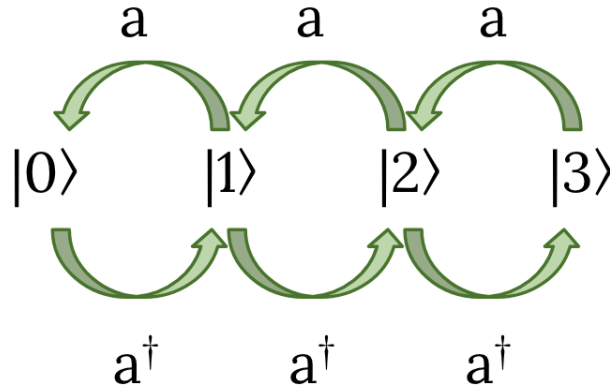


Figure 2: Energy ladder of the quantum harmonic oscillator. Levels are equally spaced by $\hbar\omega$; the vacuum carries zero-point energy $\frac{1}{2}\hbar\omega$.

1.1 Optical Phase Space and Quadrature Operators

Writing the electric field operator as $\hat{E}_x = E_0(\hat{a}e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t})\sin(kz)$, we define the *quadrature operators*

$$\hat{X}_1 = \frac{1}{2}(\hat{a} + \hat{a}^\dagger), \quad \hat{X}_2 = \frac{1}{2i}(\hat{a} - \hat{a}^\dagger), \quad (5)$$

which satisfy $[\hat{X}_1, \hat{X}_2] = \frac{i}{2}$ and hence the uncertainty relation $\Delta X_1 \Delta X_2 \geq \frac{1}{4}$. The quadratures represent the real and imaginary parts of the complex field amplitude, oscillating 90° out of phase; together they span the optical phase space.

One might expect the vacuum state $|0\rangle$ of the electromagnetic field (a mode with no photons, no current, no source) to be quiescent, with the electric field sitting exactly at zero. Instead, the non-commutativity of $\hat{a}$ and $\hat{a}^\dagger$ forbids this: a state of definite photon number $n = 0$ cannot simultaneously have a well-defined field amplitude, and the r.m.s. fluctuation of the vacuum is nonzero, at $\Delta E_x = E_0 \sin(kz) \cdot \frac{1}{\sqrt{2}}$. These *vacuum fluctuations* have observable consequences, including the Casimir effect and the Lamb shift.

2 Coherent States

2.1 Eigenstates of the Annihilation Operator

THE Fock states $|n\rangle$ are eigenstates of the photon number operator and form a natural basis, but they are poor descriptions of a classical light field: as we saw, $\langle n|\hat{E}|n\rangle = 0$ for all n , so no Fock state has a well-defined phase or amplitude. A laser beam, by contrast, behaves to excellent approximation like a classical sinusoidal wave with a definite amplitude and phase. To describe such a state quantum mechanically we need a different family of states. These will need a

HAVING nonzero field expectation value while still saturating the uncertainty bound. We can then seek them by seeking eigenstates of the (non-Hermitian) annihilation operator,

$$\hat{a}|\alpha\rangle = \alpha|\alpha\rangle, \quad \alpha \in \mathbb{C}. \quad (6)$$

Expanding in the Fock basis $|\alpha\rangle = \sum_n C_n |n\rangle$ and matching coefficients yields $C_n = e^{-|\alpha|^2/2} \alpha^n / \sqrt{n!}$, the *normalized coherent state* is

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (7)$$

The photon-number distribution is Poissonian with mean $\bar{n} = |\alpha|^2$: $P(n) = |\langle n|\alpha\rangle|^2 = e^{-\bar{n}} \bar{n}^n / n!$.

The resulting field expectation value is

$$\langle \alpha | \hat{E}_x | \alpha \rangle = 2|\alpha| \left(\frac{\hbar\omega}{2\varepsilon_0 V} \right)^{1/2} \sin(\omega t - \mathbf{k} \cdot \mathbf{r} - \theta), \quad (8)$$

with $\alpha = |\alpha|e^{i\theta}$, which is indistinguishable from a classical sinusoidal wave. The fractional photon-number uncertainty, $\Delta n / \bar{n} = 1/\sqrt{\bar{n}}$, vanishes in the classical limit $\bar{n} \rightarrow \infty$.

Coherent states simultaneously minimise the uncertainty product and distribute the uncertainty equally between the two quadratures, each carrying the vacuum noise:

$$\Delta X_1 = \Delta X_2 = \frac{1}{2}. \quad (9)$$

This equal sharing is the defining property of a *minimum-uncertainty state* and contrasts sharply with squeezed states (Section 3), where one quadrature variance is reduced below $\frac{1}{4}$.

2.2 The Displacement Operator and Displaced Vacuum

Coherent states can be simply understood as states displaced from the vacuum. Defining the *unitary displacement operator*

$$\hat{D}(\alpha) = e^{\alpha \hat{a}^\dagger - \alpha^* \hat{a}}, \quad (10)$$

one has $|\alpha\rangle = \hat{D}(\alpha)|0\rangle$. In position space the coherent-state wavefunction is a Gaussian

$$\psi_\alpha(q, t) = \left(\frac{\omega}{\pi\hbar}\right)^{1/4} \exp\left[-\frac{(\xi - \alpha e^{-i\omega t}/\sqrt{2})^2}{2} - \frac{|\alpha|^2}{2} + \frac{\xi^2}{2}\right], \quad (11)$$

where $\xi = q\sqrt{\omega/\hbar}$. Its centroid follows the classical harmonic-oscillator trajectory while the Gaussian envelope is time-independent: coherent states are *non-dispersive wave packets*.

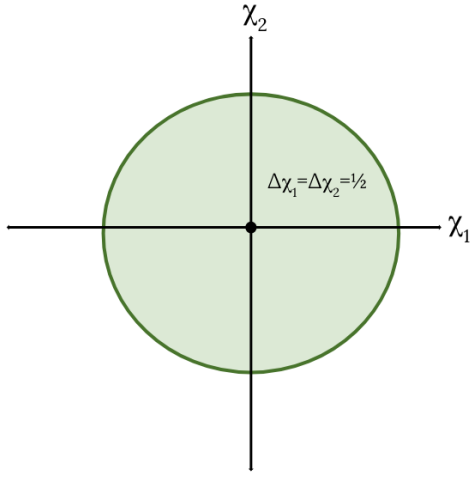


Figure 3: Vacuum state in phase space: circular error disc of radius $\frac{1}{2}$ centred at the origin, with equal noise in both quadratures.

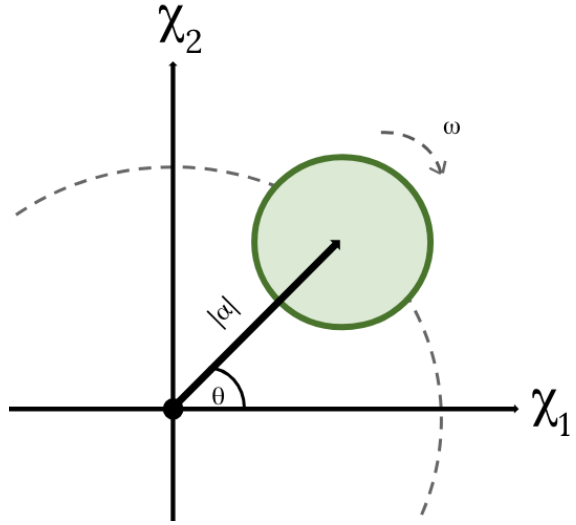


Figure 4: Coherent state $|\alpha\rangle$ displaced to $\alpha = |\alpha|e^{i\theta}$ in phase space. The error disc rotates at frequency ω , tracing a classical sinusoid.

Physically, a coherent state can be generated by coupling the vacuum to a classical oscillating current $J(\mathbf{k}, t)$, accumulating displacement $\alpha(T) = \int_0^T u(t) dt$ where $u(t) \propto e^{-i\omega t} \hat{\mathbf{e}} \cdot \mathbf{J}(\mathbf{k}, t)$.

In the complex α -plane, the quadrature expectation values $\langle \hat{X}_1 \rangle_\alpha = \text{Re}(\alpha)$ and $\langle \hat{X}_2 \rangle_\alpha = \text{Im}(\alpha)$ identify α with a point in phase space. A coherent state is represented by an error disc of radius $\frac{1}{2}$ centred at α (equal in all directions), rotating at frequency ω around the origin.

3 Squeezed States of Light

WITH this framework established, we are now equipped to understand squeezed light. Recall that for any coherent state both quadrature uncertainties saturate the minimum allowed by the commutation relation: $\Delta X_1 = \Delta X_2 = \frac{1}{2}$. A *squeezed state* redistributes this uncertainty, reducing one quadrature variance below the vacuum level at the expense of the conjugate:

$$\Delta X_1 < \frac{1}{2}, \quad \Delta X_2 > \frac{1}{2}, \quad \Delta X_1 \Delta X_2 \geq \frac{1}{4}. \quad (12)$$

Such states are generated by the *single-mode squeezing operator*

$$\hat{S}(\xi) = \exp\left[\frac{1}{2}\left(\xi^* \hat{a}^2 - \xi \hat{a}^{\dagger 2}\right)\right], \quad \xi = r e^{i\theta}, \quad (13)$$

where $r \geq 0$ is the *squeeze parameter* and $\theta \in [0, 2\pi)$ sets the squeezing angle in phase space. The operator $\hat{S}(\xi)$ is unitary, $\hat{S}^\dagger(\xi)\hat{S}(\xi) = \hat{1}$, and its action on the ladder operators follows from the Baker–Campbell–Hausdorff lemma:

$$\hat{S}^\dagger(\xi) \hat{a} \hat{S}(\xi) = \hat{a} \cosh r - \hat{a}^\dagger e^{i\theta} \sinh r, \quad (14)$$

$$\hat{S}^\dagger(\xi) \hat{a}^\dagger \hat{S}(\xi) = \hat{a}^\dagger \cosh r - \hat{a} e^{-i\theta} \sinh r. \quad (15)$$

These are the *Bogoliubov transformations* of quantum optics: a squeezed mode is a linear superposition of the original mode and its conjugate, mixing photon creation and annihilation.

3.1 Quadrature Variances

Substituting (14)–(15) into the definitions (5), the quadrature variances of the *squeezed vacuum* $|\xi\rangle = \hat{S}(\xi)|0\rangle$ are

$$\langle(\Delta\hat{X}_1)^2\rangle = \frac{1}{4}\left(\cosh^2 r + \sinh^2 r - 2 \sinh r \cosh r \cos \theta\right), \quad (16)$$

$$\langle(\Delta\hat{X}_2)^2\rangle = \frac{1}{4}\left(\cosh^2 r + \sinh^2 r + 2 \sinh r \cosh r \cos \theta\right). \quad (17)$$

For the principal squeezing axis $\theta = 0$ these simplify to

$$\langle(\Delta\hat{X}_1)^2\rangle = \frac{1}{4}e^{-2r}, \quad \langle(\Delta\hat{X}_2)^2\rangle = \frac{1}{4}e^{+2r}, \quad (18)$$

so that $\Delta X_1 \Delta X_2 = \frac{1}{4}$: the squeezed vacuum is a *minimum-uncertainty state* for all r . The semi-axes of the phase-space ellipse are $\frac{1}{2}e^{-r}$ and $\frac{1}{2}e^{+r}$, and the area is preserved at $\pi/4$.

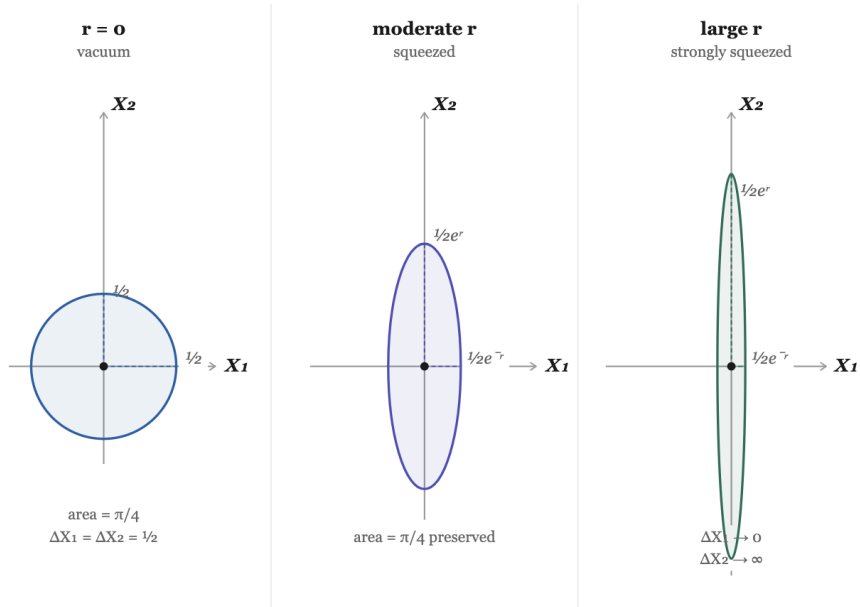


Figure 5: Phase-space error ellipse for increasing squeeze parameter r . The semi-axes $\frac{1}{2}e^{-r}$ and $\frac{1}{2}e^r$ grow and shrink reciprocally, preserving the area $\pi/4$ for all r .

For $\theta = \pi/2$ the roles of $\hat{X}_1$ and $\hat{X}_2$ are exchanged.

The degree of squeezing is conventionally quoted in decibels,

$$\mathcal{S} = -10 \log_{10}(e^{-2r}) = \frac{20r}{\ln 10} \text{ dB}, \quad (19)$$

so that $r \approx 0.115$ per dB. State-of-the-art laboratory sources have demonstrated in excess of 15 dB of squeezing [3].

3.2 Photon Statistics of the Squeezed Vacuum

The mean photon number of the squeezed vacuum is

$$\langle \hat{n} \rangle = \langle \xi | \hat{a}^\dagger \hat{a} | \xi \rangle = \sinh^2 r, \quad (20)$$

The photon-number distribution of $|\xi\rangle$ is non-zero only on even Fock states,

$$P(2k) = \frac{(2k)!}{2^{2k}(k!)^2} \frac{\tanh^{2k} r}{\cosh r}, \quad k = 0, 1, 2, \dots, \quad (21)$$

and $P(2k+1) = 0$ for all k . This *even-parity* structure reflects the fact that $\hat{S}(\xi)$ creates photons in correlated pairs via the $\hat{a}^{\dagger 2}$ term in the exponent of (13), so the photon number is always even.

Parity conservation.

We can understand the even-parity structure of the squeezed vacuum through a clearer lens: the parity operator $\hat{\Pi} = e^{i\pi \hat{a}^\dagger \hat{a}}$ commutes with the squeezing operator,

$$[\hat{S}(r), \hat{\Pi}] = 0,$$

since $\hat{S}(r)$ is generated by $\hat{a}^{\dagger 2}$ and $\hat{a}^2$, both of which change photon number by exactly two and therefore preserve parity. The vacuum is itself an even-parity eigenstate, $\hat{\Pi}|0\rangle = +|0\rangle$, so the squeezed vacuum $|\xi\rangle = \hat{S}(\xi)|0\rangle$ inherits this eigenvalue exactly: parity is a conserved quantity under squeezing.

The Fock expansion. The squeezed vacuum is an infinite superposition of even-photon-number states,

$$|\xi\rangle = \frac{1}{\sqrt{\cosh r}} \sum_{n=0}^{\infty} \sqrt{\frac{(2n)!}{2^{2n}(n!)^2}} (\tanh r)^n |2n\rangle.$$

This also illustrates why quadrature space is the natural language for optical quantum states: in Fock space the squeezed vacuum is an unwieldy infinite sum, whereas in phase space it is simply an ellipse.

The variance $\langle (\Delta \hat{n})^2 \rangle = 2 \sinh^2 r \cosh^2 r$ is super-Poissonian: squeezing reduces noise in a *quadrature*, which is a phase-sensitive quantity, whereas photon number is phase-blind. A state can therefore have sub-vacuum phase noise while simultaneously exhibiting large photon-number fluctuations.

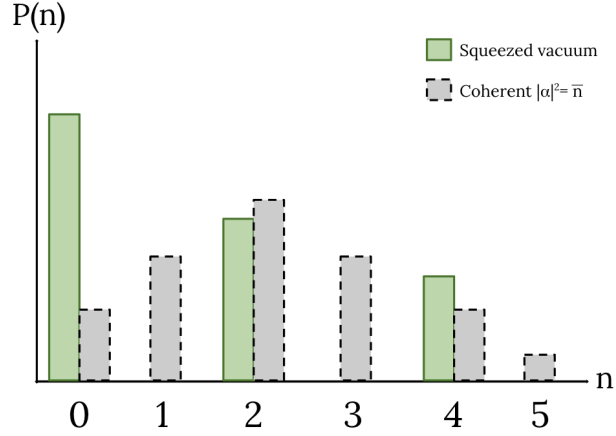


Figure 6: Photon-number distribution of a coherent state vs squeezed state with $\bar{n} = 5$. The fractional uncertainty $\Delta n/\bar{n} = 1/\sqrt{\bar{n}}$ vanishes as $\bar{n} \rightarrow \infty$.

4 Nonlinear Optics and the $\chi^{(2)}$ Interaction

HAVING established the formalism for squeezed states, we now turn to their physical generation: the $\chi^{(2)}$ nonlinear interaction that drives the pair-creation process underlying the squeezing operator (13).

In linear optics, the response of a dielectric medium to an applied electric field is encoded in the susceptibility $\chi^{(1)}$, which determines the refractive index and accounts for dispersion and absorption. However, for sufficiently intense fields, the polarization response of the medium becomes nonlinear. Expanding the induced polarization density as a power series in the electric field,

$$P_i = \epsilon_0 \left[\chi_{ij}^{(1)} E_j + \chi_{ijk}^{(2)} E_j E_k + \chi_{ijkl}^{(3)} E_j E_k E_l + \dots \right], \quad (22)$$

where repeated indices are summed (Einstein convention). The tensor $\chi_{ijk}^{(2)}$ is the *second-order (nonlinear) susceptibility*, a rank-3 tensor with $3^3 = 27$ components, and it governs processes such as second-harmonic generation (SHG), sum- and difference-frequency generation, and optical parametric amplification (OPA).

4.1 Inversion Symmetry

A centrosymmetric crystal possesses an *inversion centre*: the physical properties of the medium are unchanged under the parity transformation $\mathbf{r} \rightarrow -\mathbf{r}$. Under inversion, all polar vectors reverse sign, so the electric field transforms as

$$\mathbf{E} \longrightarrow -\mathbf{E}, \quad (23)$$

and correspondingly

$$\mathbf{P} \longrightarrow -\mathbf{P}. \quad (24)$$

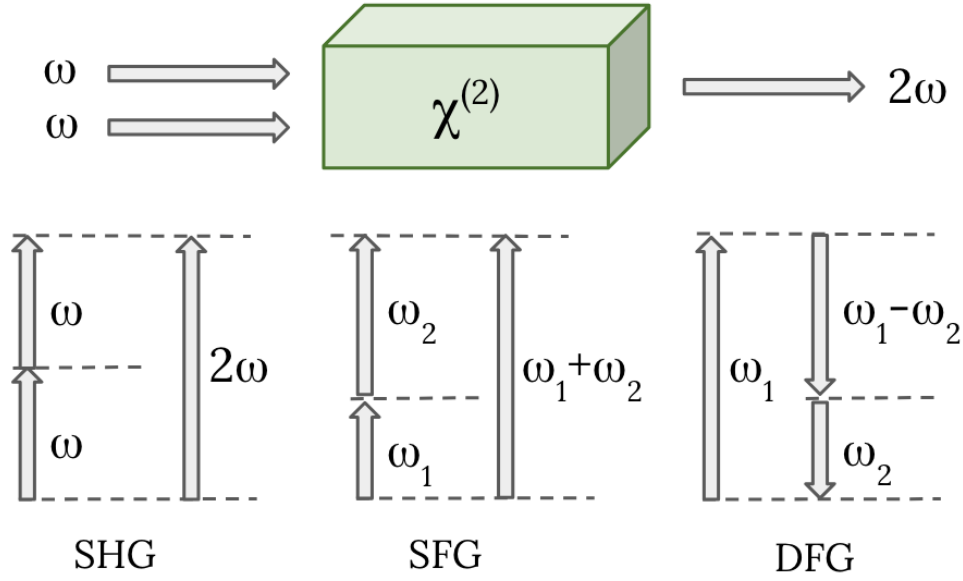


Figure 7: The three elementary $\chi^{(2)}$ processes shown as a crystal schematic (top) and energy-level diagrams (bottom). *Left* (SHG): two photons at ω are annihilated and a single photon at 2ω is created; the process underlying frequency doubling of, e.g., a Nd:YAG pump laser. *Centre* (SFG): photons at ω_1 and ω_2 combine to produce a photon at $\omega_1 + \omega_2$; used in upconversion detection and UV generation. *Right* (DFG): a pump photon at ω_1 is annihilated, stimulating emission of a signal photon at ω_2 and generating an idler at $\omega_1 - \omega_2$; this is the microscopic picture of optical parametric amplification.

Now consider the second-order contribution alone:

$$P_i^{(2)} = \epsilon_0 \chi_{ijk}^{(2)} E_j E_k. \quad (25)$$

Under inversion the left-hand side becomes $-P_i^{(2)}$, while the right-hand side picks up *two* sign changes from $E_j \rightarrow -E_j$ and $E_k \rightarrow -E_k$:

$$-P_i^{(2)} = \epsilon_0 \chi_{ijk}^{(2)} (-E_j)(-E_k) = \epsilon_0 \chi_{ijk}^{(2)} E_j E_k = P_i^{(2)}. \quad (26)$$

The only number satisfying $-x = x$ is $x = 0$, so we must have

$$\boxed{\chi_{ijk}^{(2)} = 0 \quad \forall i, j, k} \quad (27)$$

in *every* centrosymmetric medium. This is a symmetry identity: it holds exactly, to all perturbative orders and regardless of the field strength. Therefore, for the OPA process, we make use of non-centrosymmetric materials exclusively.

Example Crystals

- **KTP** (potassium titanyl phosphate, KTiOPO_4) belongs to point group $mm2$ (C_{2v}). The absence of inversion symmetry, combined with two mirror planes and a twofold axis, leaves several non-zero $\chi^{(2)}$ components, the largest being $d_{33} \approx 16$ pm/V. KTP is widely used for

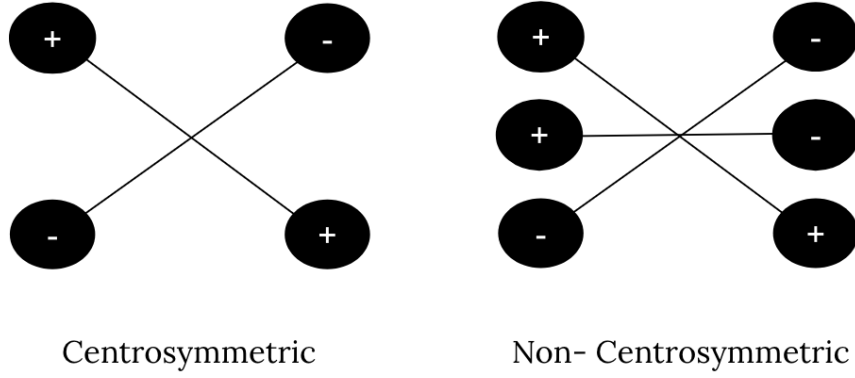


Figure 8: Physical origin of the $\chi^{(2)} = 0$ selection rule. In a centrosymmetric lattice every atom has an inversion-equivalent partner; the nonlinear polarisation induced by one site is exactly cancelled by its partner, giving zero net $P^{(2)}$. In the noncentrosymmetric case, the electron cloud would displace asymmetrically under an applied field, leaving a nonzero $P^{(2)}$.

frequency doubling of Nd:YAG lasers ($1064 \rightarrow 532$ nm).

- **BBO** (β -barium borate, β - BaB_2O_4) belongs to point group $3m$ (C_{3v}). Its relatively large d_{22} component and high damage threshold make it the material of choice for UV and deep-UV generation, as well as for ultrashort-pulse OPA/OPCPA systems.

4.2 The Nonlinear Wave Equation

The symmetry argument of the preceding section tells us *which* materials can support a $\chi^{(2)}$ interaction, but it says nothing about what that interaction actually *does* to the fields propagating through such a crystal. To answer that question we return to Maxwell's equations.

Recall from Section 1 that Maxwell's equations in the absence of free charges give $\nabla \times \mathbf{B} = \mu_0 \partial_t \mathbf{D}$, where $\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$. Separating the polarization into its linear and nonlinear parts, $\mathbf{P} = \mathbf{P}^{(1)} + \mathbf{P}^{\text{NL}}$, and using $\mathbf{P}^{(1)} = \varepsilon_0 \chi^{(1)} \mathbf{E}$, the wave equation for the electric field becomes

$$\nabla^2 \mathbf{E} - \frac{n^2}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{1}{\varepsilon_0 c^2} \frac{\partial^2 \mathbf{P}^{\text{NL}}}{\partial t^2}, \quad (28)$$

where $n = \sqrt{1 + \chi^{(1)}}$ is the linear refractive index.

The structure is that of a *driven harmonic oscillator*: the left-hand side is the free-propagation of a wave in a linear medium of index n , and the right-hand side is the forcing term. Now let's recall what the $\chi^{(2)}$ term in the polarization expansion $\mathbf{P}^{\text{NL}}$ looks like:

$$P_i^{(2)} = \varepsilon_0 \chi_{ijk}^{(2)} E_j E_k \quad (29)$$

Crucially, this term is quadratic in the electric field, so it contains products of field amplitudes. This means it oscillates at *sum and difference frequencies* not present in either input alone. For

a $\chi^{(2)}$ medium with two input fields at ω_1 and ω_2 , the source term $\chi^{(2)}E_1E_2$ drives new waves at $\omega_1 \pm \omega_2$, provided the geometry allows those waves to build up coherently.

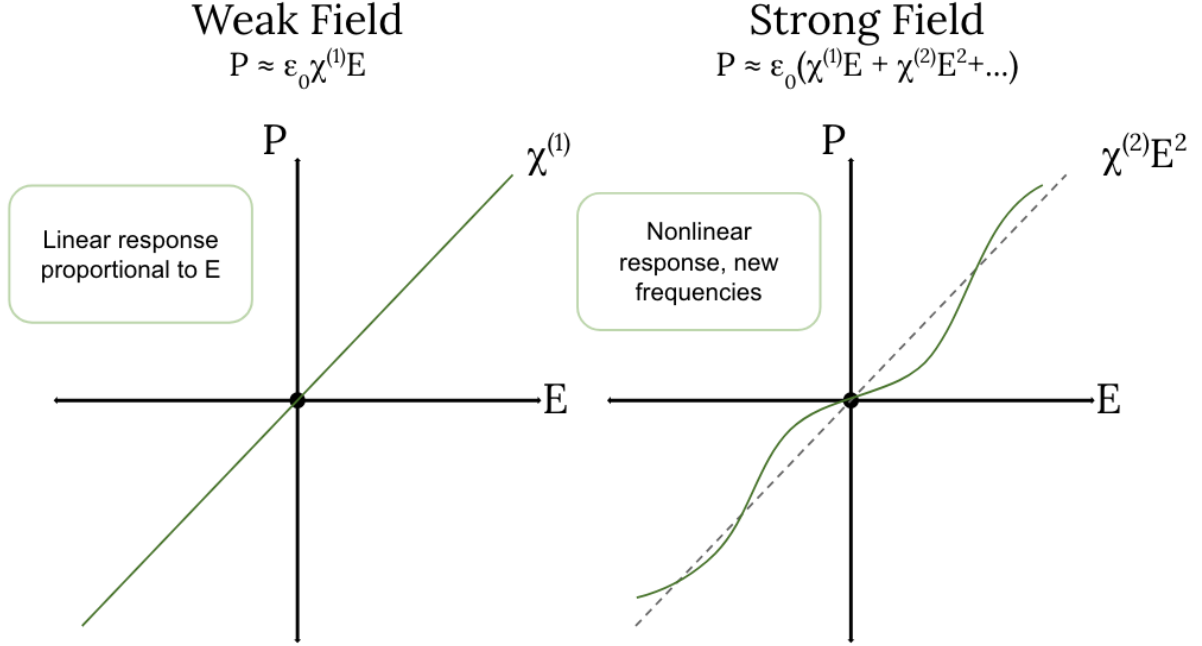


Figure 9: Polarization response $P(E)$ in weak and strong fields. At low intensity (left) the response is linear, $P \approx \epsilon_0 \chi^{(1)} E$. At high intensity (right) higher-order terms bend the curve away from linearity; the $\chi^{(2)} E^2$ contribution breaks inversion symmetry and drives polarization oscillations at new frequencies $\omega_1 \pm \omega_2$ not present in the input.

At the level of individual photons this is simply energy conservation: one pump photon at ω_p is annihilated, and in its place two daughter photons appear at signal and idler frequencies ω_s and ω_i , subject to

$$\hbar\omega_p = \hbar\omega_s + \hbar\omega_i. \quad (30)$$

But energy conservation alone does not guarantee that the new field *builds up*. As the three waves propagate through the crystal each accumulates phase at its own rate $k_j = n_j\omega_j/c$, determined by the refractive index at that frequency. Whether the signal field grows or cancels depends entirely on whether its driving polarization stays in step with the wave it is trying to excite, and that is a question of *momentum* conservation.

4.3 Coupled-Wave Equations

To extract quantitative predictions from (28) we consider three co-propagating plane waves: a *pump* at ω_p , *signal* at ω_s , and *idler* at ω_i , with $\omega_p = \omega_s + \omega_i$ imposed by the frequency content of $\chi^{(2)}E^2$. Each field is written with a slowly-varying complex envelope $\mathcal{E}_j(z)$,

$$E_j(z, t) = \frac{1}{2} \mathcal{E}_j(z) e^{i(k_j z - \omega_j t)} + \text{c.c.}, \quad (31)$$

where $k_j = n_j \omega_j / c$. The envelope $\mathcal{E}_j(z)$ is permitted to vary with z as the waves exchange energy, but the *slowly-varying envelope approximation* (SVEA) requires this variation to be gentle:

$$\left| \frac{\partial^2 \mathcal{E}_j}{\partial z^2} \right| \ll \left| k_j \frac{\partial \mathcal{E}_j}{\partial z} \right|. \quad (32)$$

Physically, we are assuming the energy exchange between the three waves is weak enough that on the scale of one optical cycle the fields look like plane waves; only on the longer scale of the crystal length does the envelope matter.

Substituting (31) into (28), invoking (32) to discard the second z -derivative of each envelope, the second-order PDE reduces to three coupled first-order equations:

$$\frac{d\mathcal{E}_s}{dz} = \frac{i\omega_s d_{\text{eff}}}{n_s c} \mathcal{E}_p \mathcal{E}_i^* e^{i\Delta k z}, \quad (33)$$

$$\frac{d\mathcal{E}_i}{dz} = \frac{i\omega_i d_{\text{eff}}}{n_i c} \mathcal{E}_p \mathcal{E}_s^* e^{i\Delta k z}, \quad (34)$$

$$\frac{d\mathcal{E}_p}{dz} = \frac{i\omega_p d_{\text{eff}}}{n_p c} \mathcal{E}_s \mathcal{E}_i e^{-i\Delta k z}, \quad (35)$$

where $d_{\text{eff}} = \chi^{(2)}/2$ is the effective nonlinear coefficient and

$$\Delta k = k_p - k_s - k_i \quad (36)$$

is the *phase mismatch*. These equations show us that the rate at which the signal grows is proportional to the product of the pump and idler amplitudes, modulated by the phase factor $e^{i\Delta k z}$ that accumulates as the waves travel. It is this phase factor that governs whether the three-wave interaction adds up constructively or averages to zero.

4.4 Phase Matching

To understand the impact of the phase factor, $e^{i\Delta k z}$, we can think of each infinitesimal slab dz of crystal as contributing a small phasor $d\mathcal{E}_s \propto e^{i\Delta k z} dz$ to the growing signal field. If $\Delta k = 0$, every slab points in the same direction in the complex plane and the contributions add coherently: the signal amplitude grows linearly with crystal length l_X . If $\Delta k \neq 0$, successive phasors rotate at rate Δk , and after a *coherence length*

$$l_c = \frac{\pi}{\Delta k}, \quad (37)$$

they have accumulated a phase of π and new contributions subtract from those already built up. The net signal amplitude integrated across the crystal is

$$\mathcal{E}_s(l_X) \propto \int_0^{l_X} e^{i\Delta k z} dz = l_X \text{sinc}\left(\frac{\Delta k l_X}{2}\right) e^{i\Delta k l_X/2}, \quad (38)$$

so the power conversion scales as $\text{sinc}^2(\Delta k l_X/2)$, peaking sharply at $\Delta k = 0$ and falling off over a crystal length $l_X \sim 2\pi/\Delta k$.

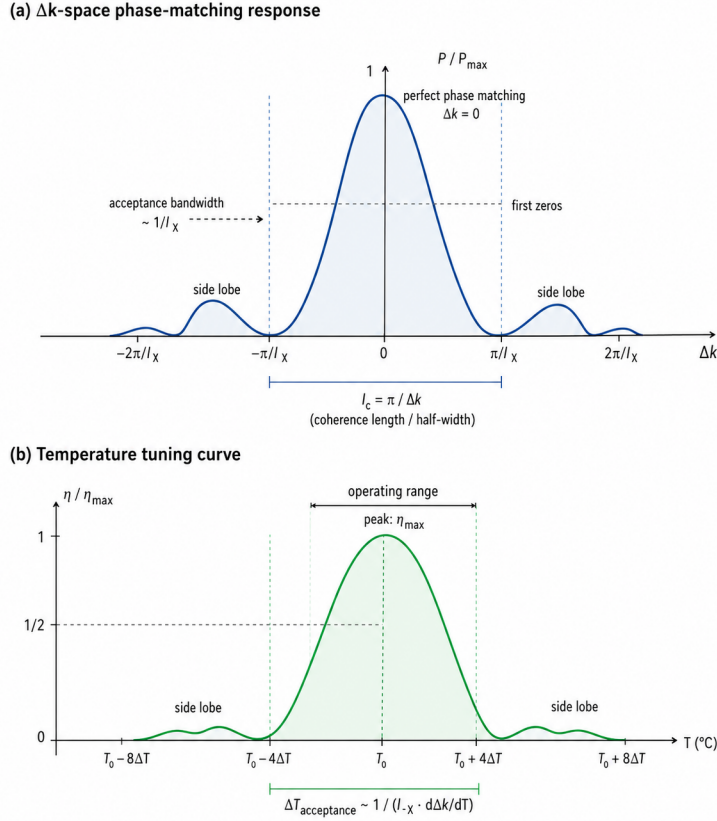


Figure 10: (a) Power conversion $\propto \text{sinc}^2(\Delta k l_X/2)$ versus phase mismatch; the acceptance bandwidth scales as $1/l_X$. (b) Efficiency versus crystal temperature, peaking at the phase-matching temperature T_0 ; the thermal acceptance $\Delta T \sim 1/(l_X \cdot d\Delta k/dT)$ sets the oven stability requirement.

Maximum conversion therefore demands the *phase-matching condition*,

$$k_p = k_s + k_i, \quad \omega_p = \omega_s + \omega_i, \quad (39)$$

which is the photon-momentum analogue of energy conservation: both energy *and* momentum must be conserved by the interacting triplet. Since $k_j = n_j \omega_j / c$, this constrains the refractive indices at all three frequencies simultaneously.

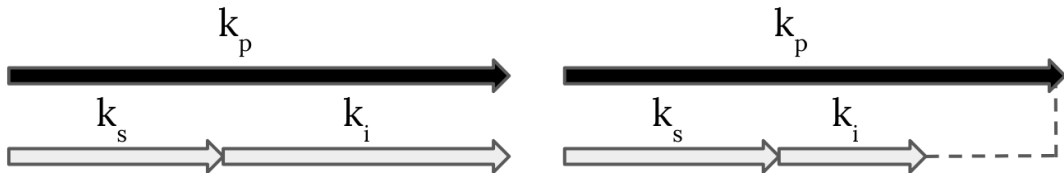


Figure 11: Momentum conservation for the OPA three-wave interaction. *Left:* phase-matched case $\Delta k = 0$, where $k_p = k_s + k_i$ and the signal builds up without limit. *Right:* mismatched case $\Delta k \neq 0$; the shortfall (dashed) means successive contributions walk out of phase and the signal cannot grow coherently.

In a normally dispersive medium $n(\omega)$ increases with frequency, so $n_p > n_s, n_i$ and $\Delta k > 0$, so

phase matching is not automatically satisfied and must be carefully engineered.

Birefringent phase matching. One way to satisfy this phase matching condition is exploiting the fact that anisotropic crystals support two refractive indices (ordinary and extraordinary), and by choosing the polarization and propagation angle one can arrange $n_p^{(e)}(\theta) = n_s^{(o)}$ at some particular temperature.

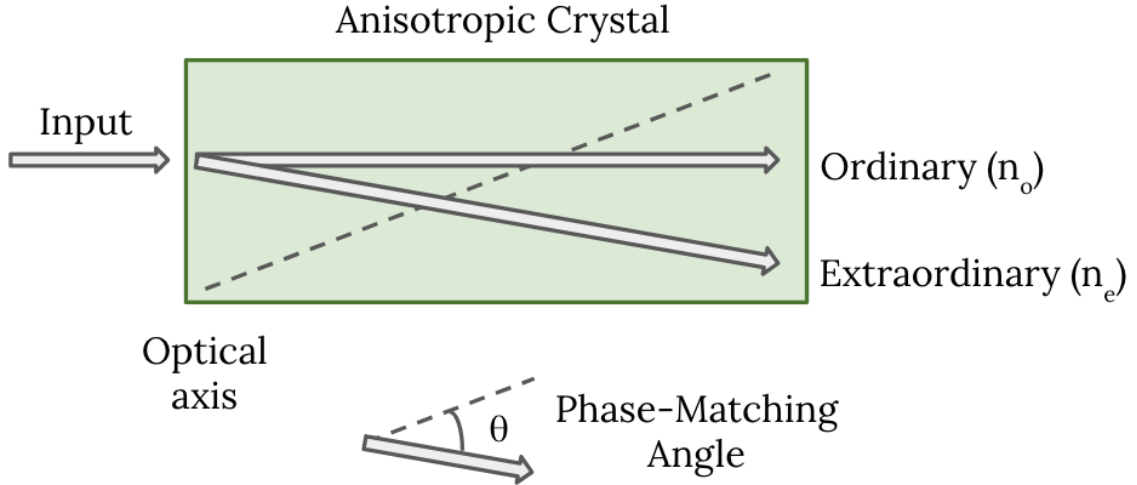


Figure 12: Birefringent phase matching. The input beam splits into ordinary (n_o) and extraordinary ($n_e(\theta)$) rays inside the anisotropic crystal. Tilting the crystal to the phase-matching angle θ relative to the optical axis arranges $\Delta k = 0$, at the cost of Poynting-vector walk-off between the two rays.

The difficulty is that the phase-matching angle is rarely the angle that maximises d_{eff} : in lithium niobate, birefringent phase matching forces operation along $d_{31} \approx 4.3 \text{ pm/V}$ rather than the principal axis $d_{33} \approx 27 \text{ pm/V}$. [5]. A further penalty is *Poynting-vector walk-off*: the ordinary and extraordinary beams propagate at a small angle to one another, reducing the effective overlap length for tightly focused beams.

Quasi-phase matching and periodic poling. We can employ a different strategy to avoid these penalties entirely. Rather than eliminating Δk , quasi-phase matching (QPM) prevents the back-transfer of energy by periodically reversing the sign of the coupling before it can undo the work already done. Recall that after one coherence length the signal and its driving polarization are π out of phase and energy would begin to flow back to the pump. If instead the sign of d_{eff} is flipped at that moment, the new contributions once again add constructively and the signal continues to grow. Repeating this reversal every l_c sustains unidirectional energy transfer throughout the entire crystal.

Physically, this sign reversal is achieved by *periodic poling*: applying a strong electric field to a ferroelectric crystal such as KTiOPO_4 inverts the spontaneous polarization of selected domains, reversing the sign of $\chi^{(2)}$ with spatial period $\Lambda = 2l_c$. In the coupled equations, this periodic modulation introduces a reciprocal lattice vector $k_0 = 2\pi/\Lambda$, replacing the phase mismatch condition

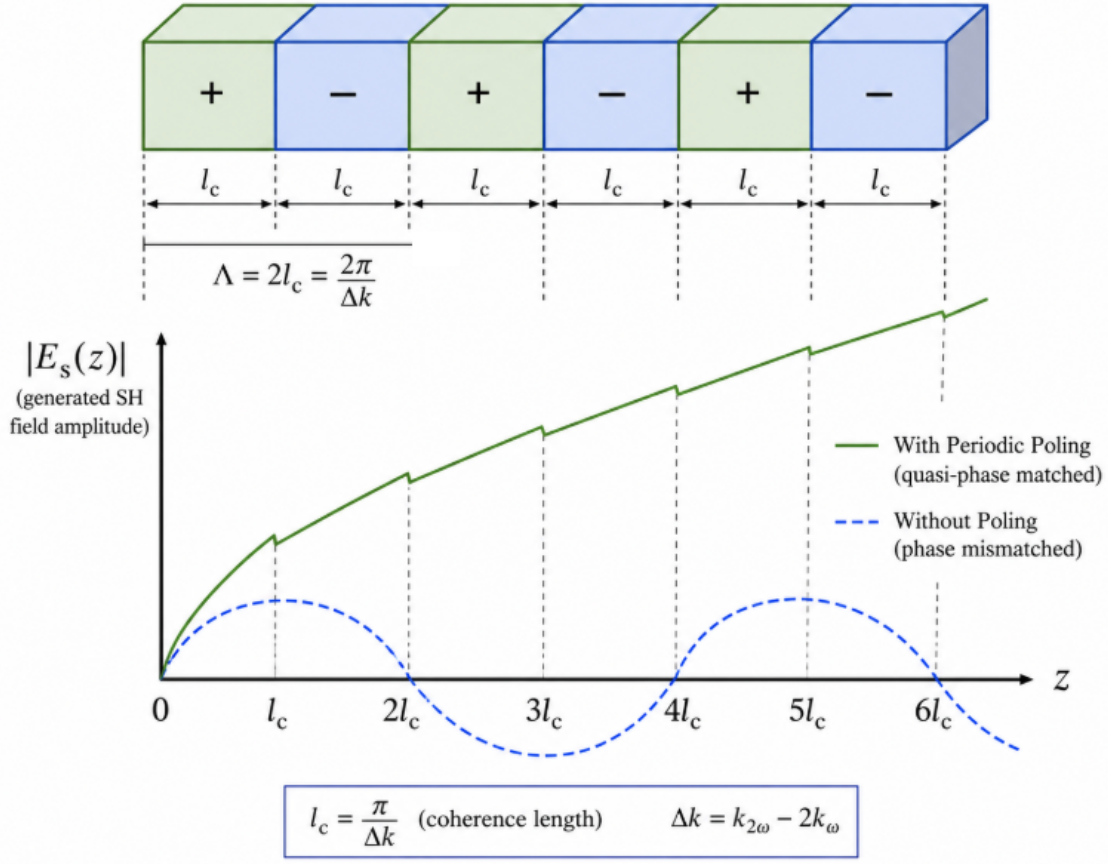


Figure 13: Quasi-phase matching via periodic poling. *Top:* A periodically-poled crystal with alternating domain orientations of period $\Lambda = 2l_c = 2\pi/\Delta k$. *Bottom:* Signal field amplitude $|E_s(z)|$ versus propagation distance z . Without poling (dashed blue) the amplitude oscillates sinusoidally with period $2l_c$ and yields no net growth. With periodic poling (solid green) the sign flip at each coherence length redirects the back-transfer, sustaining monotonic growth throughout the crystal. The definitions $l_c = \pi/\Delta k$ and $\Delta k = k_{2\omega} - 2k_\omega$ are given for the second-harmonic case; for OPA the analogous condition is $\Delta k = k_p - k_s - k_i$.

(39) with

$$\Delta k_{\text{eff}} = k_p - k_s - k_i - k_0 = 0, \quad (40)$$

where Λ is simply chosen to satisfy this relation at the desired wavelengths and temperature. The resulting periodically-poled crystal supports efficient parametric interaction without birefringent walk-off and with an effective nonlinear coefficient

$$d_{\text{eff}}^{\text{QPM}} = \frac{2}{\pi} d_{33}, \quad (41)$$

where the factor $2/\pi$ is the first-order Fourier coefficient of the square-wave domain pattern [6]. This modest reduction from the bulk d_{33} is more than compensated by the freedom to exploit the largest tensor element of the material, the elimination of walk-off, and the ability to engineer Λ for any desired interaction wavelength by design rather than by finding a special propagation angle.

For the $532\text{ nm} \rightarrow 1064\text{ nm}$ interaction, a ppKTP crystal with period $\Lambda \approx 9\text{--}10\text{ }\mu\text{m}$ satisfies (40) near room temperature. The poling period Λ is fixed at fabrication and compensates the natural

phase mismatch only at a single design temperature T_0 ; the consequences of temperature drift are examined in Section 5.

5 Focused Gaussian Beams and Phase-Matching Corrections

THE coupled-wave equations of Section 4.3 were derived assuming infinite plane-wave beams. In practice the pump is a focused Gaussian, and focusing introduces two corrections to the phase-matching picture developed above. First, the finite beam cross-section reduces the effective interaction volume relative to a plane wave, so the conversion efficiency depends on how tightly the beam is focused inside the crystal. Second, and more subtly, a focused Gaussian accumulates a longitudinal phase shift — the Gouy phase — that modifies the effective wavevector and shifts the true phase-matching condition away from $\Delta k = 0$. Boyd and Kleinman [8] showed that both effects are captured by a single dimensionless function $h(\sigma, \xi, B)$ that multiplies the plane-wave conversion efficiency.

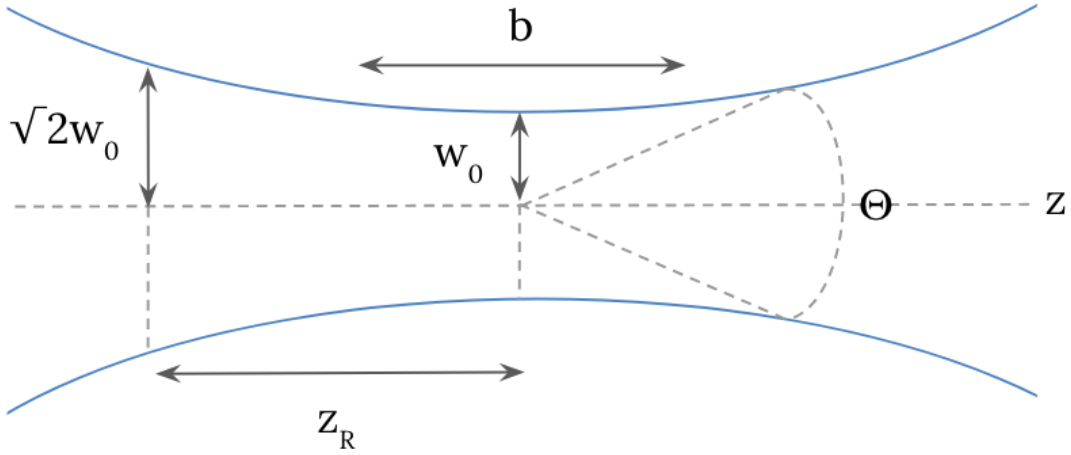


Figure 14: Geometry of a focused Gaussian beam. The beam waist w_0 occurs at the focus; at the Rayleigh range $z_R = \pi w_0^2/\lambda$ the radius has grown to $\sqrt{2}w_0$. The confocal parameter $b = 2z_R = k_1 w_0^2$ sets the natural length scale for focusing inside the crystal; the Boyd–Kleinman parameter $\xi = l_X/b$ counts how many confocal lengths fit within the crystal. In the far field the beam diverges at half-angle $\Theta = \lambda/(\pi w_0)$.

5.1 The Focusing Function $h(\xi, \sigma, B)$

Relative to plane waves, for a gaussian beam the second-harmonic (or parametric) conversion power acquires a dimensionless correction factor h [8]:

$$P_{\text{SHG}} = \frac{2\omega_1^2 d_{\text{eff}}^2 l_X k_1}{\pi \epsilon_0 c^3 n_1^2 n_2} \cdot P_1^2 \cdot h_{\text{SHG}}(\sigma, \xi, B), \quad (42)$$

where P_1 is the fundamental power, $k_1 = n_1 \omega_1/c$ is the fundamental wavenumber, and the focusing function is the double integral

$$h(\sigma, \xi, B) = \frac{1}{4\xi} \int_{-\xi}^{+\xi} d\tau_1 \int_{-\xi}^{+\xi} d\tau_2 \frac{e^{i\sigma(\tau_1 - \tau_2)} \cdot e^{-B^2(\tau_1 - \tau_2)^2/\xi}}{(1 + i\tau_1)(1 - i\tau_2)}. \quad (43)$$

The arguments encode the three physical effects competing inside the crystal. The *focusing parameter* $\xi = l_X/b$, where $b = w_0^2 k_1$ is the confocal parameter, measures how many Rayleigh lengths fit inside the crystal: $\xi \ll 1$ is loose focusing (long z_R , gentle divergence) and $\xi \gg 1$ is tight focusing (short z_R , rapid divergence past the waist).

The *phase-mismatch parameter* $\sigma = b\Delta k/2$ measures the accumulated phase walk-off scaled by the confocal length; for perfect quasi-phase matching $\Delta k = 0$ so $\sigma = 0$, but the optimum of h occurs at a small positive σ as discussed below. The *walk-off parameter* $B = \rho\sqrt{l_X k_1}/2$ encodes the Poynting-vector walk-off angle ρ ; for a quasi-phase-matched crystal such as ppKTP operating along the z -axis, $\rho = 0$ and $B = 0$, which simplifies the integral considerably.

5.2 The Optimum and the $\xi^* = 2.84$ Result

Figure 15 shows the surface $h(\sigma, \xi)$ for $B = 0$. The function rises steeply from zero at small ξ (the loose-focusing limit, where the beam diverges too rapidly to sustain a high intensity throughout the crystal) and falls gently at large ξ (the tight-focusing limit, where the Gouy phase $\phi_G(z) = \arctan(z/z_R)$ accumulates rapidly and successive crystal slabs contribute with alternating signs, degrading the coherent build-up. The peak is shallow enough that operating just off the maxima at $\xi \approx 1\text{--}2$ costs only $\sim 20\text{--}30\%$ of the gain.

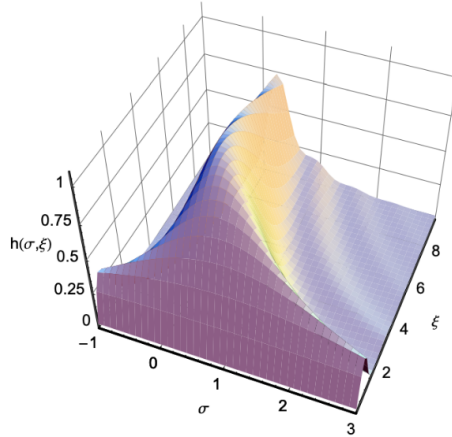


Figure 15: The Boyd–Kleinman focusing function $h(\sigma, \xi)$ evaluated at $B = 0$ (zero walk-off, appropriate for ppKTP). The surface has a unique global maximum of $h^* = 1.068$ at $(\xi^*, \sigma^*) = (2.84, \sigma_{\text{opt}} > 0)$. Evaluating at $\sigma = 0$ gives a spurious subsidiary peak near $\xi \approx 1.4$ with $h \approx 0.64$, which is not the true optimum; the joint optimisation over both ξ and σ is essential.

A critical consequence of the Gouy phase is its effect on the σ dependence of h . Because the Gouy phase shifts the effective wavevector of the focused beam by an amount $\sim 1/z_R$, the phase matching condition $\Delta k_{\text{eff}} = 0$ is satisfied not at the nominal $\Delta k = 0$ but at a small positive Δk , i.e. at $\sigma > 0$.

Evaluating the integral via joint maximisation

$$(\xi^*, \sigma^*) = \arg \max_{\xi, \sigma} h(\sigma, \xi, 0), \quad (44)$$

gives $\xi^* = 2.84$, $\sigma^* \approx 0.573$, and $h^* = 1.068$ [8].

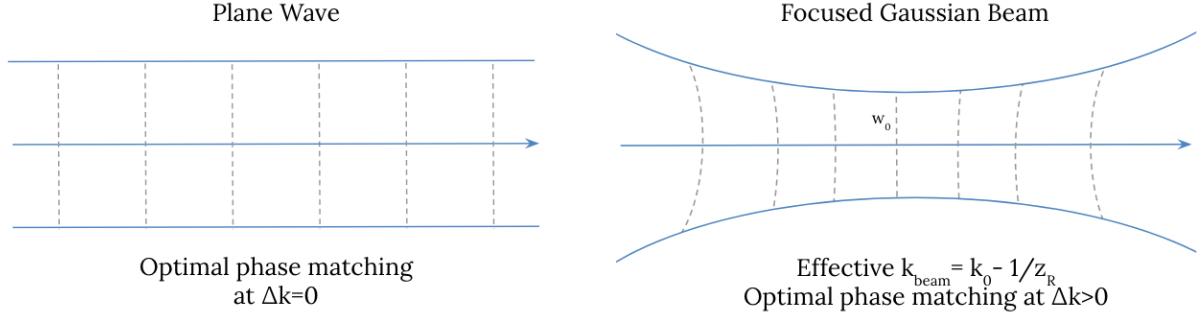


Figure 16: A focused Gaussian acquires a longitudinal Gouy phase $\phi_G(z) = \arctan(z/z_R)$, reducing the effective wavevector to $k_{\text{beam}} \approx k_0 - 1/z_R$. To compensate, the crystal poling must satisfy $\Delta k > 0$, i.e. $\sigma > 0$, shifting the true optimum of $h(\sigma, \xi)$ away from the $\sigma = 0$ axis.

Numerical evaluation of $h(\xi, \sigma)$. For $B = 0$ the double integral (43) reduces to a product of two real one-dimensional integrals. Separating the real and imaginary parts of the integrand $e^{i\sigma\tau}/(1+i\tau)$ explicitly:

$$I_{\text{re}} = \int_{-\xi}^{+\xi} \frac{\cos(\sigma\tau) + \tau \sin(\sigma\tau)}{1 + \tau^2} d\tau,$$

$$I_{\text{im}} = \int_{-\xi}^{+\xi} \frac{\sin(\sigma\tau) - \tau \cos(\sigma\tau)}{1 + \tau^2} d\tau,$$

so that $h(\xi, \sigma) = (I_{\text{re}}^2 + I_{\text{im}}^2)/4\xi$. This reduces the problem to two standard one-dimensional integrals, each smooth on the bound $[-\xi, \xi]$. The maximum of h must then be found by optimising jointly over (ξ, σ) .

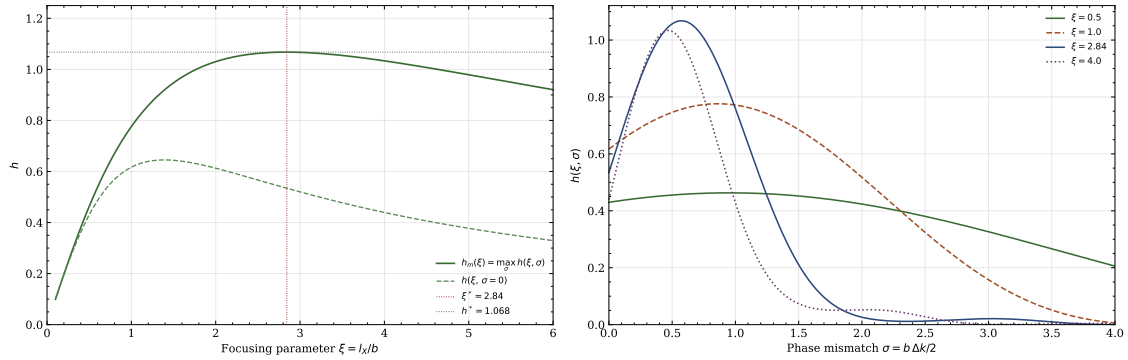


Figure 17: Two-dimensional slices of $h(\sigma, \xi)$. *Left:* $h_m(\xi) \equiv \max_{\sigma} h(\sigma, \xi)$ (solid green) peaks at $\xi^* = 2.84$, $h^* = 1.068$ (dotted lines), while $h(\xi, \sigma = 0)$ (dashed green) peaks spuriously at $\xi \approx 1.4$. *Right:* $h(\xi, \sigma)$ versus σ for four values of ξ . Tighter focusing ($\xi = 4.0$, dotted purple) demands precise phase matching and has a narrow acceptance bandwidth; loose focusing ($\xi = 0.5$, green) is very forgiving. The $\xi = 2.84$ curve (blue) achieves the highest peak while retaining a reasonable bandwidth.

5.3 Optimal Beam Waist as a Function of Crystal Length

Given the optimal focusing parameter ξ^* , the corresponding beam waist follows immediately from the definition $\xi = l_X/b = l_X \lambda / (2\pi n w_0^2)$:

$$w_0^*(\xi) = \left(\frac{l_X \lambda_{\text{pump}}}{2\pi n_{\text{crystal}} \xi} \right)^{1/2}, \quad (45)$$

where $\lambda_{\text{pump}} = 532 \text{ nm}$ is the green pump wavelength and n_{crystal} is the ppKTP refractive index at that wavelength ($n \approx 1.889$ at 532 nm , from the Sellmeier equation for KTP).

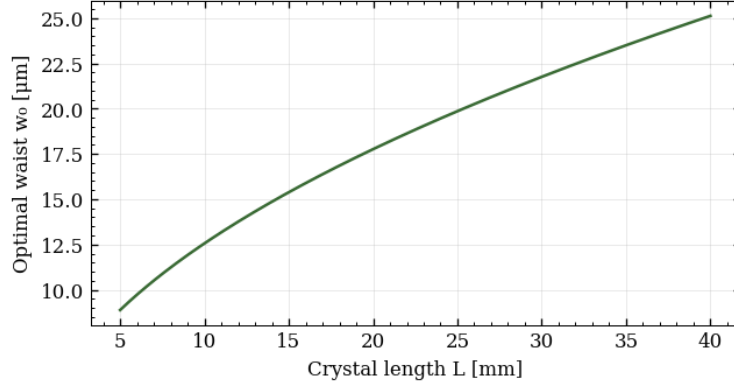


Figure 18: Optimal 532 nm pump beam waist w_0^* versus crystal length l_X from Eq. (45) at $\xi = 2.84$.

At $\xi = \xi^* = 2.84$ and $l_X = 10 \text{ mm}$, Eq. (45) gives $w_0^* \approx 12.9 \mu\text{m}$, while at $l_X = 20 \text{ mm}$ the optimum shifts to $w_0^* \approx 17.8 \mu\text{m}$, scaling as $\sqrt{l_X}$. The $\sqrt{l_X}$ scaling means that longer crystals naturally accommodate larger waists at the same ξ^* , reducing peak intensity while maintaining optimal overlap.

The shallow peak of $h_m(\xi)$ around $\xi^* = 2.84$ means that operating at $\xi \approx 1$ sacrifices only $\sim 20\%$ of the conversion efficiency ($h(1, \sigma_{\text{opt}}) \approx 0.84 h^*$) while substantially reducing the intracavity intensity. For a fixed crystal length, the beam waist at $\xi = 1$ is a factor of $\sqrt{2.84} \approx 1.69$ larger than at $\xi^* = 2.84$, reducing the peak intensity by nearly a factor of three.

However, since the pump power required to maintain a fixed parametric gain scales as $P_{\text{pump}} \propto w_0^2$, operating at $\xi = 1$ imposes a $\sim 4\times$ pump-power penalty relative to the BK optimum. Physically, a larger spot size dilutes the pump photons over more area, requiring more of them to achieve the same interaction strength. Conversely, for a fixed pump power, a longer crystal at the same ξ^* naturally produces a larger waist via Eq. (45), reducing peak intensity without sacrificing efficiency.

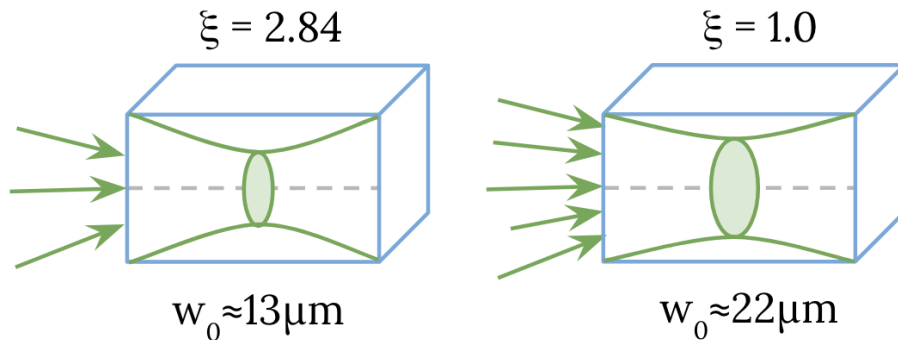


Figure 19: Comparison of tight ($\xi = 2.84$) and loose ($\xi = 1.0$) focusing inside a ppKTP crystal. At the Boyd–Kleinman optimum the beam converges to $w_0 \approx 13 \mu\text{m}$, maximising the nonlinear overlap integral $h(\xi, \sigma)$. Operating at $\xi = 1.0$ yields a larger waist $w_0 \approx 22 \mu\text{m}$, reducing the peak intensity at a cost of $\sim 4\times$ more pump power for the same parametric gain.

5.4 Temperature Dependence of Phase Matching

The poling period Λ satisfies $\Delta k_{\text{eff}} = 0$ only at the fabrication design temperature T_0 . In practice, all three refractive indices n_p , n_s , n_i drift with temperature via the material Sellmeier equations, so Δk_{eff} acquires a temperature dependence $\Delta k(T)$. The conversion efficiency therefore inherits the same sinc^2 form derived in Section 4.4, now as a function of temperature rather than spatial detuning:

$$\eta_{\text{PM}}(T) = \text{sinc}^2\left(\frac{\Delta k(T) l_X}{2}\right), \quad (46)$$

where $\Delta k(T)$ is evaluated from the KTP Sellmeier equations of Ref. [9]. The crystal length l_X plays the same role here as propagation distance in Eq. (38): it sets the acceptance bandwidth via the same Fourier relationship.

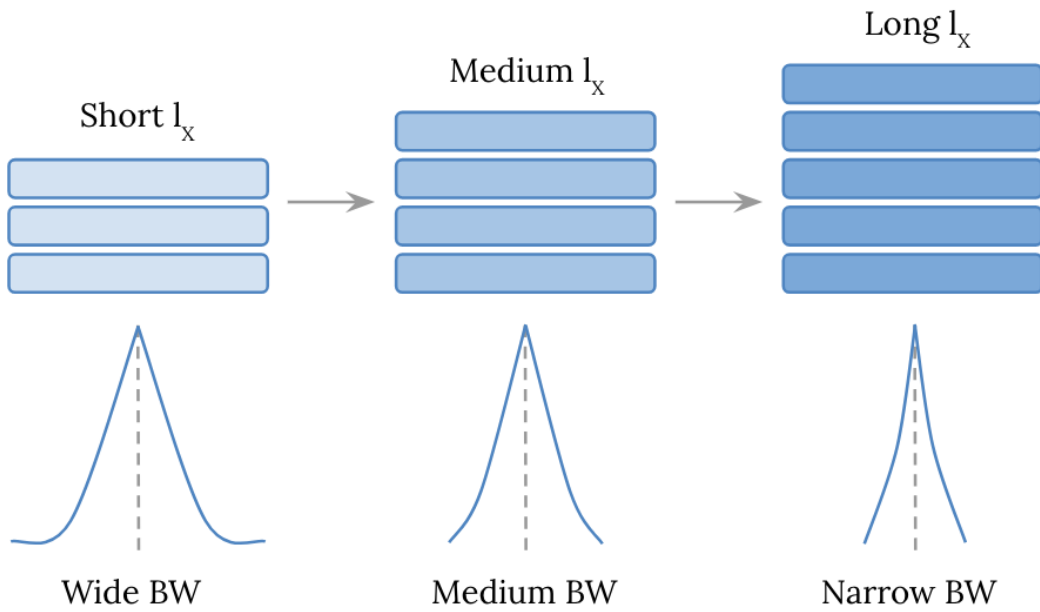


Figure 20: Crystal length as a Fourier filter. A longer crystal (more poling slices) demands phase agreement across a greater propagation distance, narrowing the thermal acceptance bandwidth as $\Delta T_{\text{FWHM}} \propto 1/l_X$.

The efficiency drops to half its peak value when $\Delta k \cdot l_X/2 = 0.4429\pi$, giving a FWHM thermal acceptance bandwidth

$$\Delta T_{\text{FWHM}} \approx \frac{1.772\pi}{l_X |d\Delta k/dT|}, \quad (47)$$

where $d\Delta k/dT$ is evaluated numerically from the KTP Sellmeier equations of Ref. [9] via a finite difference on $\Delta k(T)$.

The $1/l_X$ Bandwidth as a Fourier Relationship

To understand Fig. 20, we can think of the crystal as a filter that only "accepts" contributions from slices that are in phase with each other. The longer the crystal, the more slices we are demanding agreement from, and the more demanding we are, the narrower the range of temperatures (or wavelengths) that can all agree simultaneously. The exact same thing happens in Fourier anal-

ysis: a longer time window results in a narrower frequency resolution. A short window is sloppy about frequency; a long window is precise. The crystal length plays the role of the window, and the phase-mismatch $\Delta k(T)$ plays the role of frequency.

6 Optical Parametric Amplification

WITH the nonlinear interaction established, we now place the ppKTP crystal of Section 4 inside a resonant optical cavity to produce squeezed vacuum via optical parametric amplification. The physical mechanism is the $\chi^{(2)}$ process derived above operating in the degenerate limit $\omega_s = \omega_i = \omega_0$, $\omega_p = 2\omega_0$: the pump field at 532 nm drives correlated pair creation at 1064 nm. The pairwise nature of this creation is the microscopic origin of squeezing, and it maps directly onto the algebraic structure of the squeezing operator $\hat{S}(\xi)$ derived in Section 3.

6.1 Single-Pass Gain & Parametric Approximation

With phase matching assummed, we can now ask what the coupled equations (33)–(35) predict for a crystal driven well below the saturation regime. In the *parametric approximation*, the pump is intense enough that its depletion is negligible: $\mathcal{E}_p \approx \text{const}$. Equation (35) decouples, and (33)–(34) become a *linear* system driven by the fixed pump amplitude.

At perfect phase matching ($\Delta k = 0$) this system integrates immediately. Defining the single-pass parametric gain $\Gamma = |g|l_X/2$, where $g = \omega_s d_{\text{eff}} \mathcal{E}_p / n_s c$, the signal and idler amplitudes after propagating a crystal of length l_X are

$$\mathcal{E}_s(l_X) = \mathcal{E}_s(0) \cosh \Gamma + i e^{i\phi_p} \mathcal{E}_i^*(0) \sinh \Gamma, \quad (48)$$

$$\mathcal{E}_i(l_X) = \mathcal{E}_i(0) \cosh \Gamma + i e^{i\phi_p} \mathcal{E}_s^*(0) \sinh \Gamma, \quad (49)$$

where ϕ_p is the pump phase.

The hyperbolic functions are the signature of *parametric amplification*: unlike a simple linear amplifier, the signal grows exponentially with interaction length rather than linearly, and the process simultaneously generates an idler from vacuum in the conjugate mode.

The physical content of (48)–(49) is exactly the Bogoliubov transformation that appeared in the quantum-optical treatment of squeezing (14)–(15). This is not a coincidence: promoting the classical envelopes to operators $\hat{a}$, $\hat{b}$ in the rotating-wave approximation, the $\chi^{(2)}$ interaction Hamiltonian in the degenerate limit $\omega_s = \omega_i = \omega_0$, $\omega_p = 2\omega_0$ takes the form

$$\hat{H}_I = i\hbar\chi \left(e^{-2i\phi_p} \hat{a}^{\dagger 2} - e^{+2i\phi_p} \hat{a}^2 \right), \quad (50)$$

where $\chi \propto \chi^{(2)} |\mathcal{E}_p|$ [1]. Time evolution under $\hat{H}_I$ for a duration t generates precisely the squeezing operator $\hat{S}(\xi)$ of (13) with $r = \chi t$ and squeezing angle $\theta = 2\phi_p$. The OPA is therefore a *direct physical implementation* of $\hat{S}(\xi)$: the pump drives correlated pair creation via the $\hat{a}^{\dagger 2}$ term, the squeeze parameter r is set by the pump power and crystal length, and rotating ϕ_p rotates the squeezing angle in phase space.

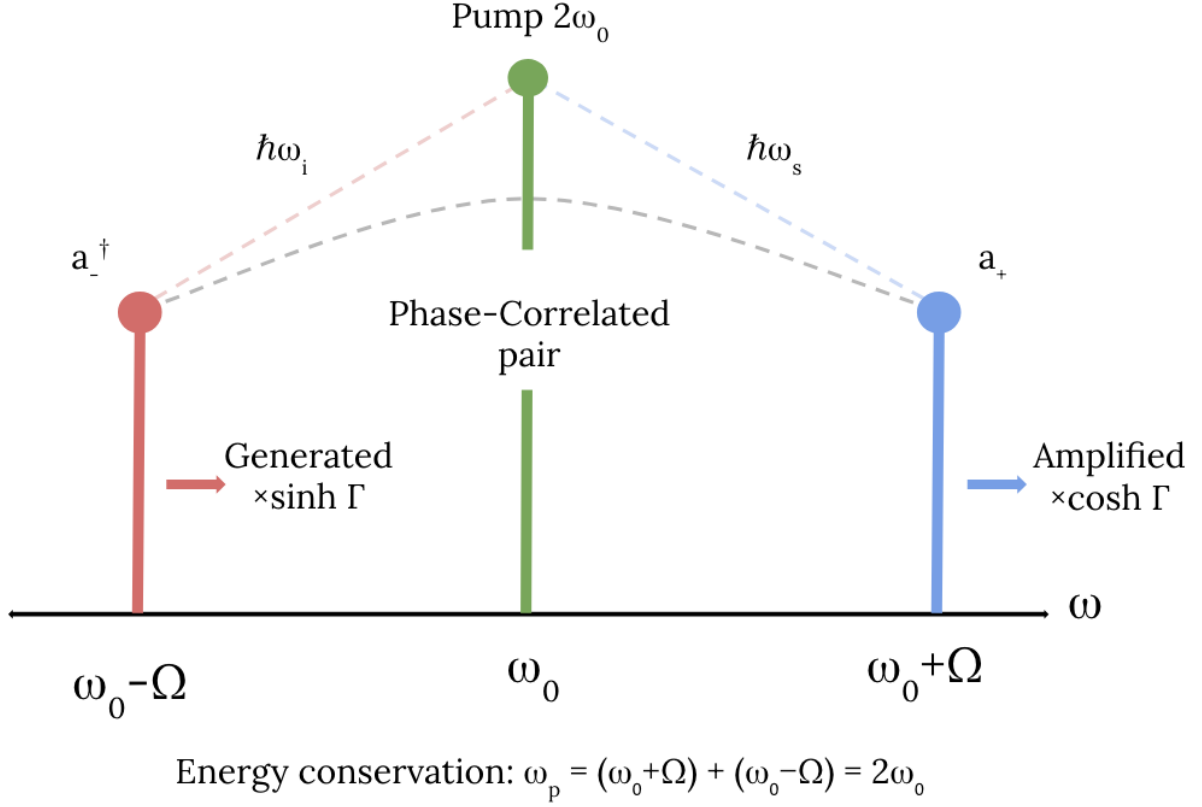


Figure 21: Parametric sideband mixing in the frequency domain. The undepleted pump at $2\omega_0$ drives correlated pair creation into the upper sideband a_+ at $\omega_0 + \Omega$ and conjugate lower sideband $a_-^\dagger$ at $\omega_0 - \Omega$, with the upper amplified by $\cosh \Gamma$ and the lower generated from vacuum by $\sinh \Gamma$.

In a single pass through a realistic crystal, Γ is modest: for a 10 mm ppKTP crystal driven at tens of milliwatts the single-pass gain is quite small. To reach the squeezing levels relevant for gravitational-wave detection, the interaction must be resonantly enhanced. This is the role of the optical cavity, which we develop in the following section.

6.2 Single-Pass Transfer Matrix

The Heisenberg equations of motion for the intracavity field follow directly from the interaction Hamiltonian in the parametric approximation. . Defining upper and lower sidebands a_+ , a_- at frequencies $\omega_0 \pm \Omega$, we collect them into the vector $\mathbf{a} = (a_+, a_-^\dagger)^T$. In a $\chi^{(2)}$ medium the pump couples these sidebands via

$$\frac{d\mathbf{a}}{dz} = \frac{1}{2} \begin{pmatrix} i\Delta k & -i\gamma \\ +i\gamma^* & -i\Delta k \end{pmatrix} \mathbf{a}(z), \quad (51)$$

where $i\gamma = i\omega_0 d_{\text{eff}} E_b / cn_a$ is the nonlinear coupling per unit length and Δk is the phase mismatch of Eq. (36) [4]. The matrix on the right-hand side of Eq. (51) belongs to the Lie algebra $\mathfrak{su}(1,1)$; consequently its matrix exponential (which gives the solution) belongs to $\text{SU}(1,1)$, and can be parametrised by a single hyperbolic amplitude and two phases.

Generators and commutation relations.

The Lie algebra $\mathfrak{su}(1, 1)$ is spanned by three generators

$$K_+ = \frac{1}{2}\hat{a}^{\dagger 2}, \quad K_- = \frac{1}{2}\hat{a}^2, \quad K_0 = \frac{1}{2}\left(\hat{n} + \frac{1}{2}\right),$$

satisfying $[K_0, K_{\pm}] = \pm K_{\pm}$ and $[K_+, K_-] = -2K_0$. K_+ creates correlated photon pairs via $\hat{a}^{\dagger 2}$; K_- destroys them; K_0 counts half the photon number and generates $U(1)$ phase rotations.

Contrast with $\mathfrak{su}(2)$. The algebra $\mathfrak{su}(2)$ has $[J_+, J_-] = +2J_0$; the only difference from $\mathfrak{su}(1, 1)$ is the sign of the right-hand side. Geometrically, $SU(2)$ acts on the sphere S^2 ($|z_1|^2 + |z_2|^2 = 1$) whose orbits are compact closed circles (a spin rotated by 2π returns to itself). The sign flip $+2J_0 \rightarrow -2K_0$ replaces the Euclidean metric with a Minkowski metric ($|z_1|^2 - |z_2|^2 = 1$), opening S^2 into the one-sheeted hyperboloid H^2 . Orbits on H^2 are open hyperbolic curves: there is no north pole, and the state never returns.

Why squeezing is motion up H^2 . The vacuum $|0\rangle$ sits at the waist of H^2 ($r = 0$). The squeezing operator

$$\hat{S}(\xi) = \exp[r(e^{-i\theta}K_+ - e^{i\theta}K_-)]$$

is a geodesic boost along H^2 : the squeeze parameter $r \geq 0$ is arc length from the waist, and the angle θ selects the direction of the boost in the (K_+, K_-) plane, i.e. the squeezing angle in phase space. Because H^2 is non-compact, r is unbounded and the quadrature variance $\langle(\Delta\hat{X}_1)^2\rangle = \frac{1}{4}e^{-2r}$ decreases monotonically without limit, unlike any $SU(2)$ rotation. The mean photon number $\langle\hat{n}\rangle = \sinh^2 r$ grows with the same arc length, reflecting the energetic cost of moving further up the sheet.

Parametrisation of H_X . Every element of $SU(1, 1)$ admits the Iwasawa decomposition $U = e^{i\phi K_0} e^{i\rho(e^{-i\theta}K_+ - e^{i\theta}K_-)} e^{i\psi K_0}$, which at perfect phase matching evaluates to

$$H_X = \begin{pmatrix} \cosh r e^{i\phi} & e^{i\psi} \sinh r \\ e^{i\psi} \sinh r & \cosh r e^{-i\phi} \end{pmatrix}, \quad \cosh^2 r - \sinh^2 r = 1.$$

The three real parameters are $r \geq 0$ (squeeze amplitude, set by pump power and crystal length), $\psi \in [0, 2\pi)$ (squeezing angle, set by pump phase $\psi = 2\phi_p$), and $\phi \in [0, 2\pi)$ (overall phase, unobservable). This is the minimum parametrisation: no more degrees of freedom exist in any $SU(1, 1)$ matrix.

To solve Eq. (51) explicitly we remove the phase mismatch by the substitution

$$\tilde{\mathbf{a}}(z) = \text{diag}(e^{+i\Delta k z/2}, e^{-i\Delta k z/2}) \mathbf{a}(z)$$

which converts the system to one with constant coefficients. Defining $\mu = \sqrt{|\gamma|^2 - (\Delta k)^2}$, the solution after propagating a crystal of length l_X is the single-pass transfer matrix

$$H_X = P(\Delta k) M(\mu, \gamma, \Delta k), \quad (52)$$

where

$$P(\Delta k) = \begin{pmatrix} e^{-i\Delta k l_X/2} & 0 \\ 0 & e^{+i\Delta k l_X/2} \end{pmatrix}$$

is the phase-ramp matrix and

$$M = \begin{pmatrix} \cosh \frac{\mu l_X}{2} + \frac{i\Delta k}{\mu} \sinh \frac{\mu l_X}{2} & -\frac{i\gamma}{\mu} \sinh \frac{\mu l_X}{2} \\ +\frac{i\gamma^*}{\mu} \sinh \frac{\mu l_X}{2} & \cosh \frac{\mu l_X}{2} - \frac{i\Delta k}{\mu} \sinh \frac{\mu l_X}{2} \end{pmatrix}$$

is the parametric transfer matrix.

At perfect phase matching, $\Delta k = 0$ and $\mu = |\gamma|$, so $\Gamma \equiv |\gamma|l_X/2$ and the transfer matrix simplifies

to

$$H_X \xrightarrow{\Delta k \rightarrow 0} \begin{pmatrix} \cosh \Gamma & e^{-2i\psi_b} \sinh \Gamma \\ e^{+2i\psi_b} \sinh \Gamma & \cosh \Gamma \end{pmatrix}, \quad (53)$$

where $\Gamma = |\gamma|l_X/2$ is the single-pass parametric gain [4].

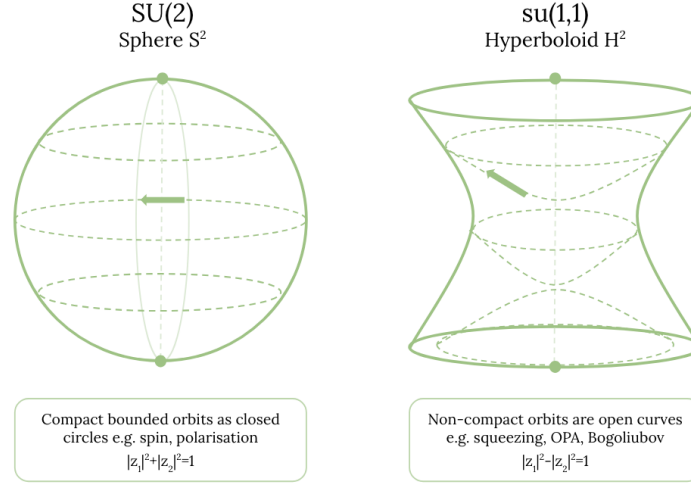


Figure 22: Comparison of the group geometries underlying the OPA transfer matrix. The squeeze parameter r measures arc length along the hyperboloid from the waist, and the transfer matrix H_X is parametrised by exactly r and two phases.

The hyperbolic structure of H_X is the sideband-basis avatar of the Bogoliubov transformations (14)–(15): the upper sideband is amplified and its conjugate lower sideband is generated, exactly as the squeezing operator mixes $\hat{a}$ and $\hat{a}^\dagger$. For $\Gamma \ll 1$, the off-diagonal elements are linear in γl_X , recovering the perturbative result implicit in Eq. (51); for $\Gamma \sim 1$ near threshold, the hyperbolic functions grow rapidly and significant squeezing is achieved in a single pass.

6.3 Resonant Enhancement and Cavity Dynamics

Single-pass gain Γ is modest for typical crystal lengths ($l_X \sim 10$ mm) and pump powers accessible in the laboratory. Enclosing the crystal in a resonant optical cavity multiplies the effective interaction by the cavity finesse, enabling threshold to be reached at pump powers of order tens of milliwatts. A natural cavity geometry for this purpose is the doubly-resonant travelling-wave (bow-tie) cavity: the four-mirror ring geometry eliminates back-reflection into the pump laser and avoids intracavity standing waves, at the cost of slightly more mirror loss than a linear cavity.

The round-trip dynamics of the intracavity sideband vector $\mathbf{a}$ are described by the self-consistency condition- after one complete round trip through the cavity, the field must reproduce itself. Physically, this is the optical analogue of the lasing threshold condition: the cavity sustains an intracavity field only when the round-trip gain exactly balances the round-trip loss.

Each mirror j with amplitude reflectivity r_j and transmissivity t_j imposes the boundary condition

$$\mathbf{a}^{(j')} = -r_j \mathbf{a}^{(j)} + t_j \mathbf{a}^{(\text{in},j)}, \quad (54)$$

and free propagation between elements contributes phase delays encoded in diagonal matrices

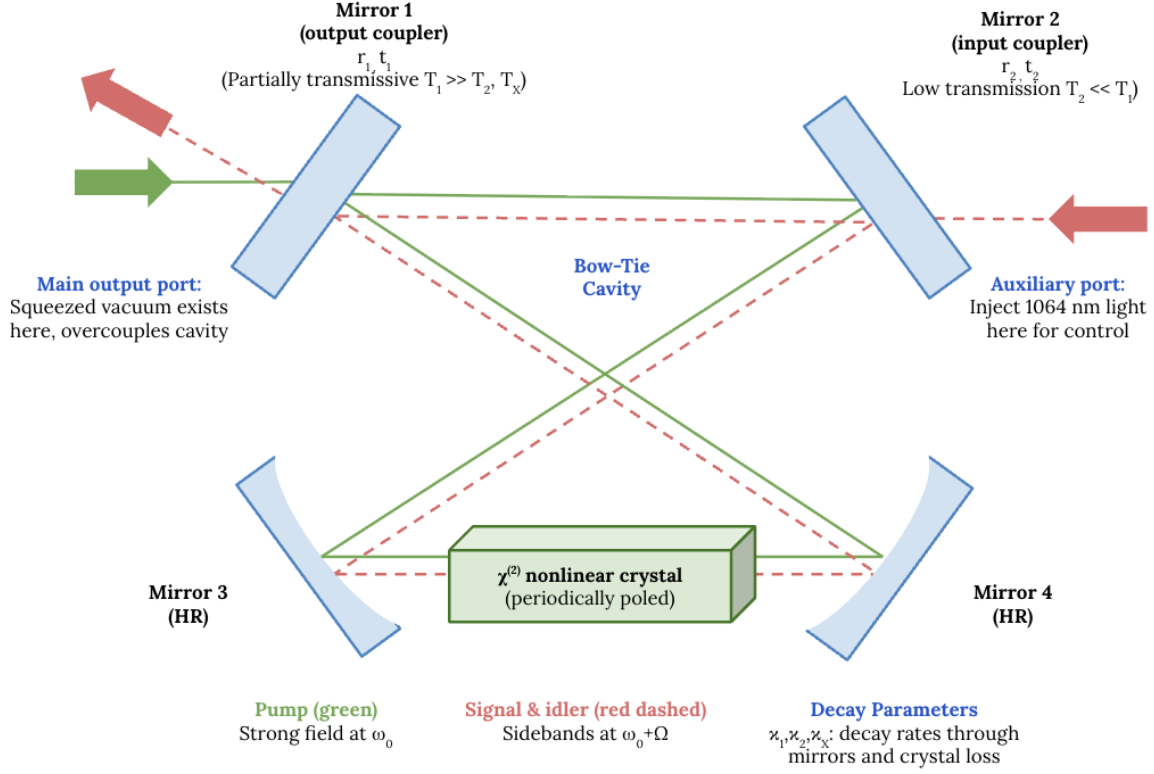


Figure 23: A doubly-resonant bow-tie OPA cavity. The travelling-wave geometry eliminates back-reflection into the pump and intracavity standing waves, and the overcoupled output mirror ensures the escape efficiency $\eta = \kappa_1/\kappa$ dominates the squeezing budget.

$L_j(s) = e^{-s\tau_j} \text{diag}(e^{-i\omega_0\tau_j}, e^{+i\omega_0\tau_j})$, where τ_j is the relevant propagation time and $s = i\Omega$ is the Laplace variable [4]. Assembling these ingredients, the intracavity field satisfies

$$\mathbf{a} = (1 - G)^{-1} \left(t_1 \mathbf{a}^{(1)} - r_1 t_2 L_{12} \mathbf{a}^{(2)} - r_1 r_2 r_X L_{12} L_{2X} \mathbf{a}^{(X)} \right), \quad (55)$$

where $G(s) = r_1 r_2 t_X e^{-s\tau_a} H_X(\dots)$ is the single round-trip gain matrix, τ_a is the round-trip time, and $\mathbf{a}^{(X)}$ represents vacuum fluctuations entering through crystal loss. The matrix $(1 - G)^{-1}$ is the *internal gain*: it diverges when G has a unit eigenvalue, which defines the OPA threshold.

The cavity is characterised by three decay rates. The dominant output coupling κ_1 (through the high-transmission mirror from which squeezed vacuum is extracted), a small secondary coupling κ_2 used for control and diagnostics, and the intrinsic crystal loss rate $\kappa_X = \mathcal{A}L_X/2\tau_a$, where $\mathcal{A}$ is the power loss per unit length due to absorption and scatter. These satisfy $\kappa = \kappa_1 + \kappa_2 + \kappa_X$ exactly [4]. The nonlinear coupling rate is $g = \Gamma/\tau_a$, and the normalised coupling

$$x = \frac{g}{\kappa} \quad (56)$$

runs from zero to unity; the OPA reaches *threshold* at $x = 1$, when gain exactly compensates all losses. In practice the device is operated at $x \lesssim 0.9$ to avoid instability while maintaining high squeezing.

The cavity enhances the single-pass interaction into an *amplified finesse*

$$\mathcal{F} = \frac{\pi}{\sqrt{1 - 2r \cosh \Gamma + r^2}} \simeq \frac{\pi/\tau_a}{\sqrt{\kappa^2 - g^2}}, \quad (57)$$

where $r = r_1 r_2 t_X$ collects all round-trip amplitude factors and the approximation holds to leading order in the decay and coupling rates [4]. As $x \rightarrow 1$ the finesse diverges and the cavity bandwidth $|s_{P-}| \simeq \kappa - g$ narrows to zero: near-threshold operation simultaneously maximises squeezing and reduces the bandwidth over which squeezing is produced, a fundamental trade-off that constrains the design of broadband squeezers for gravitational-wave detection.

6.4 Input–Output Relations and Squeezed Quadrature Variance

The observable output of the OPA is the field reflected from the overcoupled output mirror (mirror 1 in Fig. 23). In the Laplace domain, the reflected quadrature field is a superposition of three contributions: vacuum $\mathcal{D}^{(1)}$ entering externally at mirror 1 and squeezed by reflection; vacuum $\mathcal{D}^{(2)}$ entering at the control mirror and transmitted through the cavity; and vacuum $\mathcal{D}^{(X)}$ entering via crystal loss:

$$\mathcal{D}^{(1')}(s) = \mathbb{R}(s)\mathcal{D}^{(1)}(s) + \mathbb{T}(s)\mathcal{D}^{(2)}(s) + \mathbb{T}^{(X)}(s)\mathcal{D}^{(X)}(s), \quad (58)$$

where $\mathbb{R}$, $\mathbb{T}$, $\mathbb{T}^{(X)}$ are 2×2 quadrature transfer matrices [4]. The quadrature reflection matrix at dc, for pump phase $\psi_b = 0$, evaluates to

$$\mathbb{R}(0) \xrightarrow{\psi_b=0} \begin{pmatrix} \frac{\kappa - 2\kappa_1 - g}{\kappa - g} & 0 \\ 0 & \frac{\kappa - 2\kappa_1 + g}{\kappa + g} \end{pmatrix}, \quad (59)$$

showing explicitly that the cosine quadrature is amplified (denominator $\kappa - g < \kappa$) while the sine quadrature is deamplified (denominator $\kappa + g > \kappa$). The sign of $\mathbb{R}_{11}$ changes as κ_1 increases past $(\kappa - g)/2$: for a strongly overcoupled cavity, $\kappa_1 \approx \kappa$ and the reflected cosine quadrature is large and positive, directly reflecting the intracavity amplification back out.

Since all input fields are initialised in the vacuum state, the cross-spectral density of the output is

$$V^{(1'1')}(\Omega) = \mathbb{R}\mathbb{R}^\dagger + \mathbb{T}\mathbb{T}^\dagger + \mathbb{T}^{(X)}\mathbb{T}^{(X)\dagger}, \quad (60)$$

and the squeezed quadrature variance at dc evaluates to

$$V_s(0) = \frac{(r_1 - re^{-\Gamma}/r_1)^2 + (t_1 t_2)^2 + (t_1 r_2 r_X)^2}{(1 - re^{-\Gamma})^2}. \quad (61)$$

To leading order in the decay and coupling rates this simplifies to the more familiar Collett–Gardiner result [7]

$$V_s(0) \simeq 1 - \frac{4\eta x}{(1+x)^2}, \quad (62)$$

where $\eta = \kappa_1/\kappa$ is the *escape efficiency*: the fraction of intracavity squeezing that exits through the output coupler rather than being lost to κ_2 or κ_X .

Equation (62) is the central result of OPA theory. Its structure reveals the two independent limits

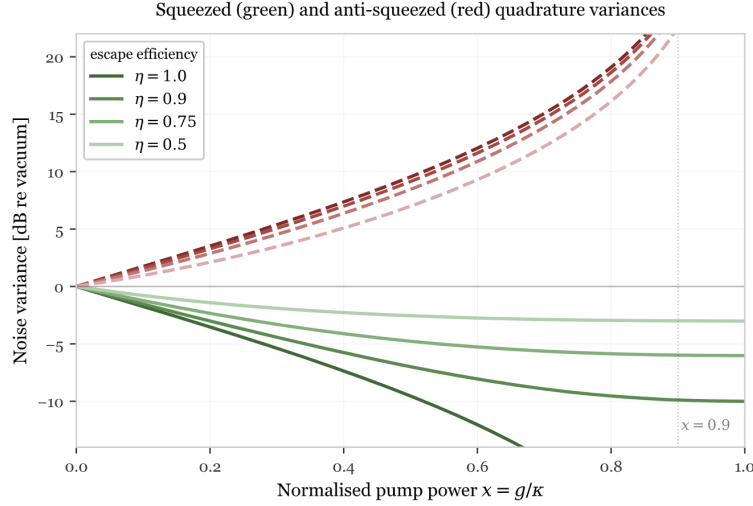


Figure 24: Quadrature variances as a function of normalised pump power $x = g/\kappa$, for several values of the escape efficiency η , from Eq. (62). As $x \rightarrow 1$ the squeezed quadrature asymptotes to the loss floor $1 - \eta$, while the anti-squeezed quadrature diverges; the dashed vertical line marks the practical operating point $x = 0.9$.

on squeezing performance. First, even at threshold ($x \rightarrow 1$) the variance approaches $1 - \eta$ rather than zero: every photon absorbed or scattered inside the cavity, or lost through the control port, mixes in a unit of unsqueezed vacuum and sets a hard floor on achievable squeezing, $\mathcal{S}_{\max} = -10 \log_{10}(1 - \eta)$ dB. Second, for fixed η and fixed x , the variance is minimised by maximising η , which requires $\kappa_1 \gg \kappa_2, \kappa_X$: the output mirror must dominate all other loss mechanisms by a wide margin.

Connecting back to Section 3, the squeezed quadrature variance maps to an effective squeeze parameter via $V_s = \frac{1}{4}e^{-2r_{\text{eff}}}$, and the degree of squeezing in decibels follows from Eq. (19). The OPA thus realises in a lossy physical system the ideal squeezing operator $\hat{S}(\xi)$ of Eq. (13): the pump drives $\hat{a}^{\dagger 2}$ pair creation, the cavity resonance amplifies the interaction via Eq. (57), and the escape efficiency η quantifies the degradation from ideal. At frequencies above the cavity bandwidth $\kappa - g$, the parametric enhancement falls off and the output returns to vacuum noise.

The practical significance of Eq. (62) becomes clear when we connect it to the sensitivity of an interferometric gravitational-wave detector. At LIGO, quantum noise limits strain sensitivity in two complementary ways: shot noise (phase uncertainty) dominates at high frequencies, while radiation-pressure noise (amplitude uncertainty) dominates at low frequencies: precisely the two conjugate quadratures of the vacuum field entering the dark port. Injecting a squeezed vacuum state with variance V_s reduces shot noise by a factor V_s relative to the vacuum level, corresponding to a strain sensitivity improvement of $\sqrt{V_s}$ in amplitude, or equivalently $-\frac{1}{2} \cdot 10 \log_{10} V_s$ dB in power.

The central message of this review is therefore that the chain from vacuum quantisation to observable sensitivity gain is unbroken: the non-commutativity of $\hat{X}_1$ and $\hat{X}_2$ sets the SQL, the squeezing operator $\hat{S}(\xi)$ circumvents it by redistributing uncertainty, and the OPA cavity realises $\hat{S}(\xi)$ physically with an efficiency set entirely by η and x .

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